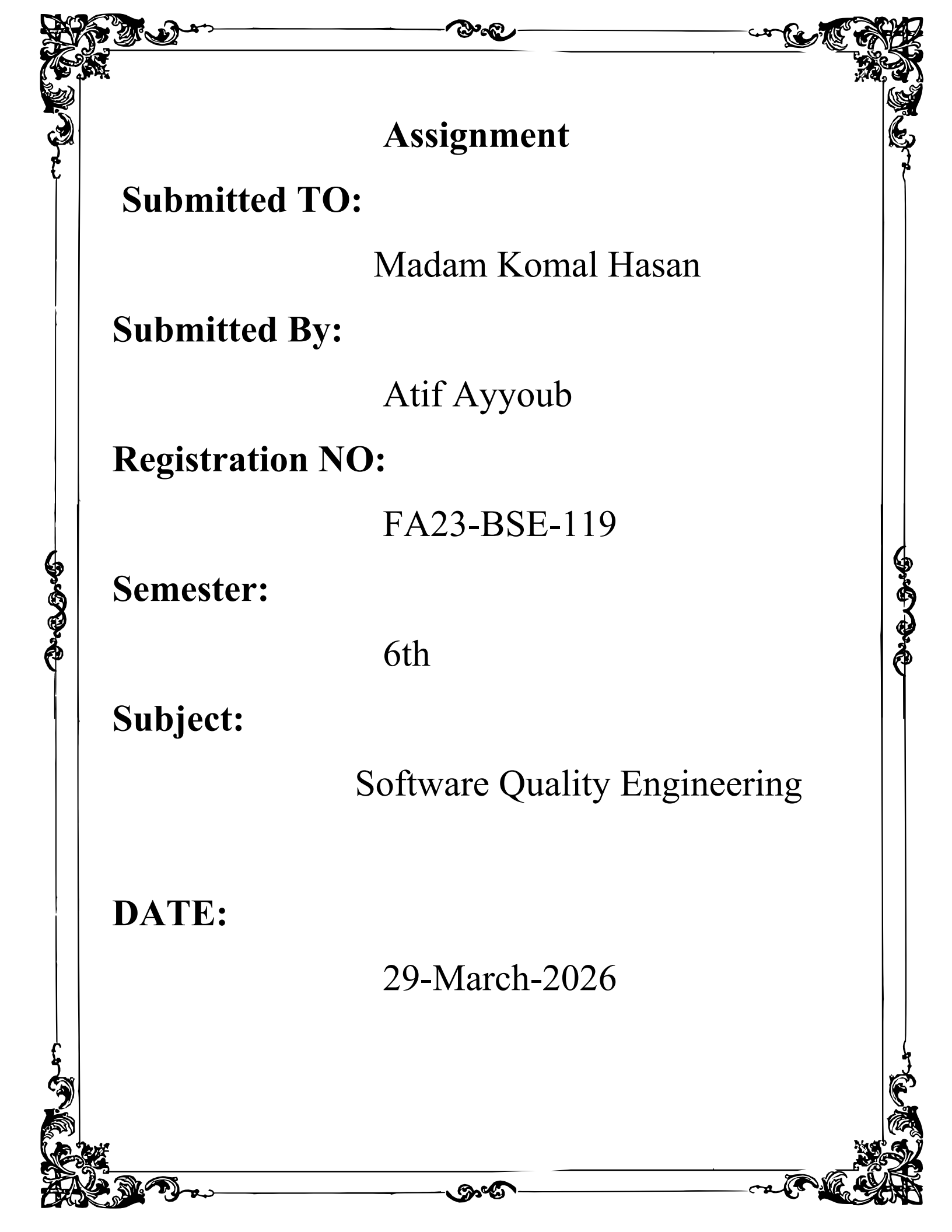




Assignment

Submitted TO:

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Submitted By:

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Registration NO:

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Semester:

6th

Subject:

Software Quality Engineering

DATE:

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Library Management System

Software Specification and Review Assignment

Prepared as a worked assignment package with walkthrough, inspection, and audit review templates

Course / Module	Software Quality Assurance / Software Engineering
Case Study	Library Management System (LMS)
Prepared For	Assignment / classroom submission / guided practice
Student Details	Name: _____ Roll No.: _____

Document purpose: This package provides a concise software specification for a Library Management System and three completed review assignments using different code artifacts. Each review uses the report format shown in the reference book and includes defects, corrective actions, preventive actions, and closure notes.

1. Assignment Brief

Objective. Prepare a quality-focused review package for a Library Management System by combining a short software specification with three review methods: walkthrough, inspection, and audit.

Expected use. The document can be submitted directly as an assignment, or used as a template where student names, signatures, and diagrams are filled later.

Part	What is included	Primary artifact	Expected outcome
A	Short software specification	Requirements + structure	Scope and design baseline
B	Walkthrough review package	Catalog search service	Defects found and improvements
C	Inspection review package	Book issue service	Detailed defect logging
D	Audit review package	Fine policy logic	Compliance gaps and closure plan

2. Software Specification - Library Management System

System purpose. The Library Management System supports book cataloging, membership management, issue/return processing, reservation handling, and fine tracking for a college or public library.

ID	Requirement	Description
FR-1	Catalog search	Users can search by title, author, ISBN, genre, and availability.
FR-2	Book issue	Librarian can issue a book only if a copy is available and member rules are satisfied.
FR-3	Book return	System records return date and triggers fine calculation when overdue.
FR-4	Reservations	Members may reserve unavailable books and receive issue alerts.
FR-5	Member management	Create, suspend, and update member profiles and borrowing limits.

FR-6	Reporting	Librarian can view overdue books, most borrowed titles, and fine summaries.
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ID	Quality attribute	Target
NFR-1	Performance	Normal catalog search should respond within 2 seconds for 50,000 books.
NFR-2	Security	Only librarians can issue, return, or modify inventory records.
NFR-3	Reliability	Borrow/return operations must be atomic and recoverable.
NFR-4	Usability	Core librarian tasks should be executable within three screens or fewer.
NFR-5	Auditability	Important transactions must be logged with user, date, and reason.

3. Architecture and Diagram Placeholders

Subsystem	Responsibility	Interfaces / notes
Catalog Service	Stores bibliographic data and search filters	Search API, index refresh, availability lookup
Loan Service	Controls issue, return, reservation, and due dates	Borrow/return commands, member checks
Fine Service	Applies overdue policy and payment updates	Fine calculator, waiver rules, receipts
Reporting Service	Generates librarian summaries and audit reports	Overdue list, activity log, trend view

Use Case Diagram Placeholder

[Insert diagram here]

Add actors: Member, Librarian, Payment Gateway; add use cases: Search Book, Issue Book, Return Book, Reserve Book, Pay Fine.

Component / Class Diagram Placeholder

[Insert diagram here]

Show Catalog, Loan, Fine, and Reporting services with Member, Book, LoanRecord, Reservation entities.

Sequence Diagram Placeholder

[Insert diagram here]

Suggested flow: Librarian -> Loan Service -> Catalog Service -> Fine Service -> Database for issue/return scenario.

4. Review Plan and Roles

Review package intent. Each review uses a different artifact so that the assignment demonstrates how review focus changes by method: walkthrough emphasizes shared understanding, inspection emphasizes defect detection discipline, and audit emphasizes process and policy conformance.

Review type	Primary role focus	Typical outputs
Walkthrough	Author-led explanation	Questions, logic corrections, maintainability improvements
Inspection	Peer defect detection	Defect log, severity reasoning, closure tracking

Audit	Compliance verification	Evidence gaps, standards mismatch, sign-off conditions
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5. Walkthrough Review Assignment

Review objective. This walkthrough reviews the Catalog Search Service. The author explains the code to the reviewers, who challenge assumptions, boundary conditions, and readability concerns.

Checklist focus. Availability logic, null handling, operator precedence, query normalization, and user-facing search behavior.

Artifact under review. A Java method used by the LMS catalog module to filter books by query string and availability.

Code Listing W-1. Catalog Search Service (Java)

```
public List<Book> searchAvailableBooks(List<Book> books, String query) {
    List<Book> result = new ArrayList<>();
    for (Book b : books) {
        if (b.getTitle().contains(query) || b.getAuthor().contains(query) && b.isAvailable()) {
            result.add(b);
        }
    }
    return result;
}
```

Observed walkthrough discussion. The logic is easy to follow, but the mixed OR/AND condition causes an availability defect. Reviewers also noted that query handling is fragile because blank or null input and case differences are not normalized.

Suggested improvement summary: add explicit parentheses, normalize title/author and query to a common case, trim inputs, reject null values safely, and add unit tests for unavailable-title matches.

Review Report

Project name: Library Management System

Name of the artifact being reviewed, with version number: Catalog Search Service v1.0

Name(s) of the reviewer(s): Student Team / Peer Review Group

Name of the author of the artifact: Module Author - Catalog Developer

Date(s) on which the review is conducted: 10-10-2026

Type of review:

☐ Independent/guided

☐ Individual/group

☒ Postal/meeting

Defects uncovered during the review (use an additional sheet if necessary)

Defect ID	Defect description	Reference to process for the defect	Defect origin	Closed on	Status (open or closed)
W-01	Title matches can be returned even when the book is unavailable because AND binds only to the author condition.	Logic	Coding	Pending	Open
W-02	No null guard for books, title, author, or query; this may trigger NullPointerException.	Robustness	Data validation	Pending	Open
W-03	Search is case sensitive and ignores leading/trailing spaces, causing inconsistent user results.	Usability	Input handling	Pending	Open
W-04	Method lacks pagination or result limits; large datasets may degrade response time.	Performance	Design	Pending	Open

Signature of lead reviewer: _____

Date of signature: _____

Closure Action by the Author

Corrective actions implemented

Action implemented	Defect IDs covered by this action	Comments
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Rewrite condition as (titleMatch authorMatch) && isAvailable	W-01	Prevents unavailable books from appearing in results
Add null checks and input normalization helper	W-02, W-03	Improves stability and search accuracy
Introduce max-result limit and paging contract	W-04	Aligns service with performance target

Preventive action implemented

Action implemented	Defect IDs covered by this action	Comments
Add code review checklist item for boolean precedence and null-safety	W-01, W-02	Reusable for service methods
Add parameterized unit tests for search edge cases	W-01, W-03, W-04	Covers unavailable, blank, and large result scenarios

Signature of author: _____

Date of signature: _____

Defect closure actions (to be filled in by the lead reviewer)

I have verified and found that all the defects described above are closed satisfactorily, except the following defects, which are retracted or pending:

1. W-04 remains pending until the API contract formally includes pagination fields.
2. All other walkthrough defects may be closed after unit test evidence is attached.
3. No retracted defects at this stage.

Signature of lead reviewer: _____

Date of signature: _____

6. Inspection Review Assignment

Review objective. This inspection reviews the Book Issue Service. The code is examined line by line using a checklist, and defects are logged without debating redesign during the detection phase.

Checklist focus. Availability checks, transaction safety, member rule enforcement, state consistency, and traceability of updates.

Artifact under review. A Python function used when a librarian issues a book to a member.

Code Listing I-1. Book Issue Service (Python)

```
def issue_book(member, book, today):
    if member.fines > 0:
        print("warning only")
    if book.copies >= 0:
        book.copies -= 1
        member.borrowed.append(book.id)
        book.last_issue_date = today
        return True
    return False
```

Observed inspection finding pattern. The function mutates multiple objects without a transaction boundary. It also permits issue when copies equal zero and does not stop members with outstanding fines or excessive loans.

Suggested improvement summary: enforce rule checks first, require copies > 0, wrap state changes in a transaction, record audit logs, and return structured errors instead of printing warnings.

Review Report

Project name: Library Management System

Name of the artifact being reviewed, with version number: Book Issue Service v1.2

Name(s) of the reviewer(s): Inspection Team / Moderator / Reader

Name of the author of the artifact: Loan Module Author

Date(s) on which the review is conducted: 12-10-2026

Type of review:

☐ Independent/guided

☒ Individual/group

☐ Postal/meeting

Defects uncovered during the review (use an additional sheet if necessary)

Defect ID	Defect description	Reference to process for the defect	Defect origin	Closed on	Status (open or closed)
I-01	Condition book.copies >= 0 allows issuing when no copy is available, resulting in negative stock.	Logic	Coding	Pending	Open
I-02	Members with unpaid fines are not blocked; function only prints a warning.	Business rule	Requirements	Pending	Open
I-03	Multiple updates are performed without transaction control, so partial updates may persist after failure.	Reliability	Design	Pending	Open
I-04	No maximum-loan check or audit log is recorded for the issue action.	Compliance	Process omission	Pending	Open

Closure Action by the Author

Corrective actions implemented

Action implemented	Defect IDs covered by this action	Comments
Change availability condition to copies > 0 and reject empty inventory	I-01	Stops negative stock
Block issuing for members with pending fines or loan-limit breach	I-02, I-04	Aligns code with stated rules
Wrap inventory/member updates and logging inside one transaction	I-03, I-04	Ensures atomicity and traceability

Preventive action implemented

Action implemented	Defect IDs covered by this action	Comments
Inspection checklist updated with inventory boundary and transaction checks	I-01, I-03	Prevents recurrence in command handlers
Introduce service contract returning success flag + reason code	I-02, I-04	Removes console-only warnings and improves UI feedback

Signature of author: _____

Date of signature: _____

Defect closure actions (to be filled in by the lead reviewer)

I have verified and found that all the defects described above are closed satisfactorily, except the following defects, which are retracted or pending:

1. I-04 remains pending until audit log schema and retention policy are approved.
2. I-01 through I-03 can be closed after transaction-level test evidence is attached.
3. No defect was rejected during inspection.

Signature of lead reviewer: _____

Date of signature: _____

7. Audit Review Assignment

Review objective. This audit verifies whether the Fine Calculation artifact conforms to documented policy, approved thresholds, and evidence expectations. The focus is not only code correctness but also policy traceability.

Checklist focus. Policy mapping, overdue boundaries, special-book handling, documentation evidence, and sign-off readiness.

Artifact under review. A JavaScript function used by the LMS Fine Service to compute overdue amount.

Code Listing A-1. Fine Calculation Logic (JavaScript)

```
function calculateFine(daysLate, bookType, memberType) {  
  let rate = 2;  
  if (bookType === "reference") rate = 0;  
  if (memberType === "faculty") rate = 1;  
  if (daysLate > 30) return rate * daysLate;  
  return rate * (daysLate - 1);  
}
```

Observed audit conclusion. The code is short, but it does not clearly implement an approved library policy. Zero or negative values are possible, the grace period is implicit rather than documented, and special handling for reference books is inconsistent with the rule that such books should not be issued externally.

Suggested improvement summary: trace each rule to a signed policy document, reject invalid negative inputs, explicitly encode grace days and caps, and retain evidence of versioned policy approval.

Review Report

Project name: Library Management System

Name of the artifact being reviewed, with version number: Fine Calculation Logic v0.9

Name(s) of the reviewer(s): Audit Reviewer / QA Representative

Name of the author of the artifact: Fine Service Author

Date(s) on which the review is conducted: 14-10-2026

Type of review:

☒ Independent/guided

☐ Individual/group

☐ Postal/meeting

Defects uncovered during the review (use an additional sheet if necessary)

Defect ID	Defect description	Reference to process for the defect	Defect origin	Closed on	Status (open or closed)
A-01	For daysLate = 0 the function returns a negative fine, which violates financial rules.	Policy logic	Coding	Pending	Open
A-02	Reference books are assigned zero rate rather than being blocked by issue policy; logic is not traceable to approved rules.	Policy compliance	Requirements	Pending	Open
A-03	Faculty rate override is hard coded without evidence of approval or version reference.	Auditability	Process gap	Pending	Open
A-04	No cap, grace-period constant, or validation for invalid input values is documented or tested.	Standards	Documentation gap	Pending	Open

Signature of lead reviewer: _____

Date of signature: _____

Closure Action by the Author

Corrective actions implemented

Action implemented	Defect IDs covered by this action	Comments
Reject negative/zero-late cases early and return zero fine explicitly	A-01	Prevents invalid financial output
Replace hard-coded rules with policy constants mapped to approved document version	A-02, A-03, A-04	Restores traceability
Add configuration for grace days, rate table, and maximum fine cap	A-02, A-04	Supports policy change without code rewrite

Preventive action implemented

Action implemented	Defect IDs covered by this action	Comments
Maintain signed policy-to-code traceability matrix for fine rules	A-02, A-03, A-04	Needed for future audits
Require audit checklist evidence: policy version, test report, and approval note	A-01 to A-04	Makes release review repeatable

Signature of author: _____

Date of signature: _____

Defect closure actions (to be filled in by the lead reviewer)

I have verified and found that all the defects described above are closed satisfactorily, except the following defects, which are retracted or pending:

1. A-03 cannot close until the approved faculty-rate memo or policy version is attached.
2. A-02 and A-04 remain conditionally open until traceability matrix is reviewed by QA.
3. A-01 can close after negative-input test cases pass and finance output is revalidated.

Signature of lead reviewer: _____

Date of signature: _____

8. Consolidated Improvement Plan

Cross-cutting lessons. Across all three review methods, the strongest themes are explicit rule encoding, boundary validation, traceability, and automated test evidence.

Improvement area	Why it matters in LMS	Recommended action
Input validation	Search, issue, and fine logic all fail when edge cases are ignored.	Adopt a shared validation utility and boundary test suite.
Transaction safety	Borrow/return activities update multiple records that must remain consistent.	Wrap critical state changes in one atomic service layer transaction.
Traceability	Audit closure depends on evidence that code implements approved policy.	Maintain requirement-to-code-to-test traceability for every release.
Review discipline	Different review types uncover different defect classes.	Use walkthrough early, inspection before coding sign-off, and audit before release.

9. Submission Checklist

- ☐ Add student name, roll number, and course details on the title page.
- ☐ Insert actual diagrams in the three placeholders if your instructor requires UML visuals.
- ☐ Optionally replace reviewer and signature lines with your own project team details.
- ☐ If the assignment must be purely template-based, clear the example defect rows and keep the structure.